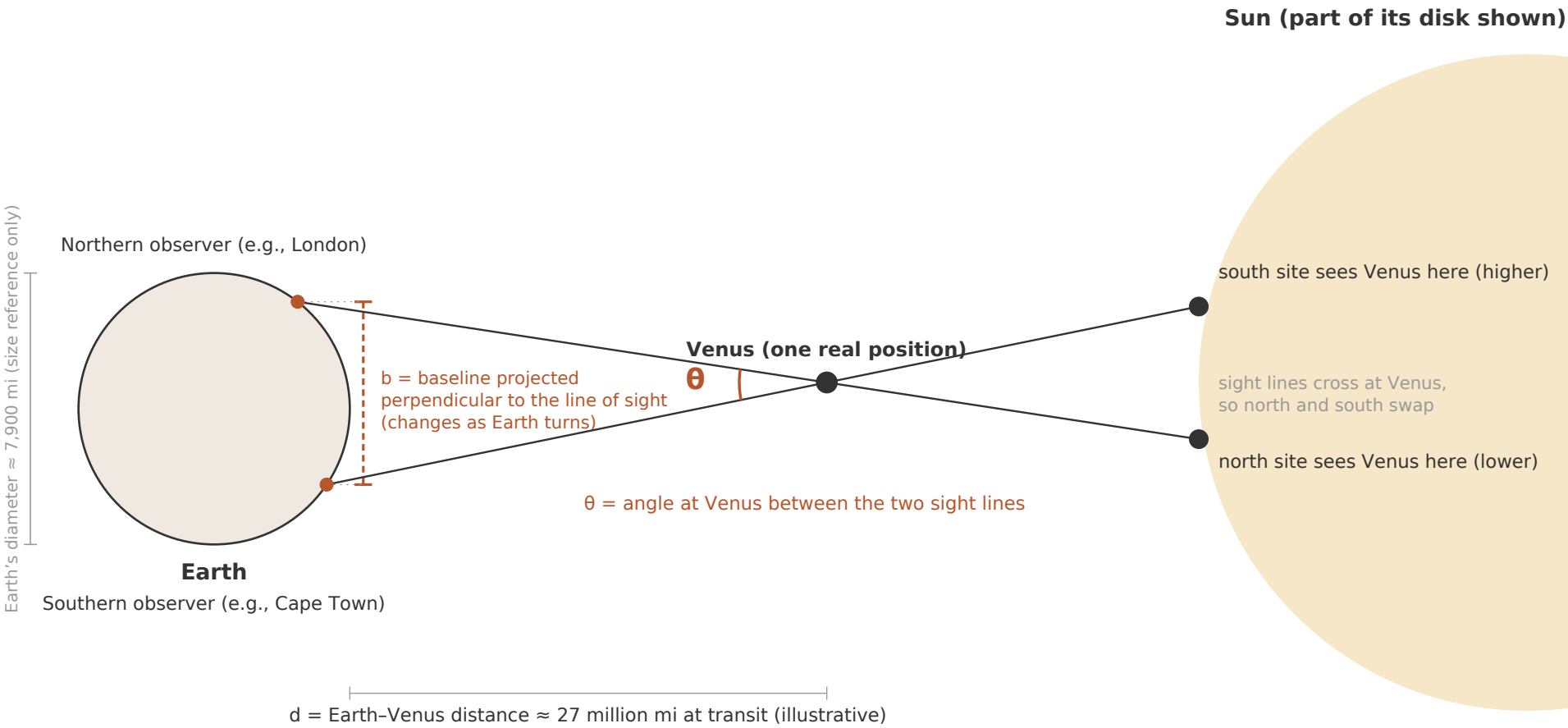
Measuring the Sun’s distance with the 1761 transit of Venus

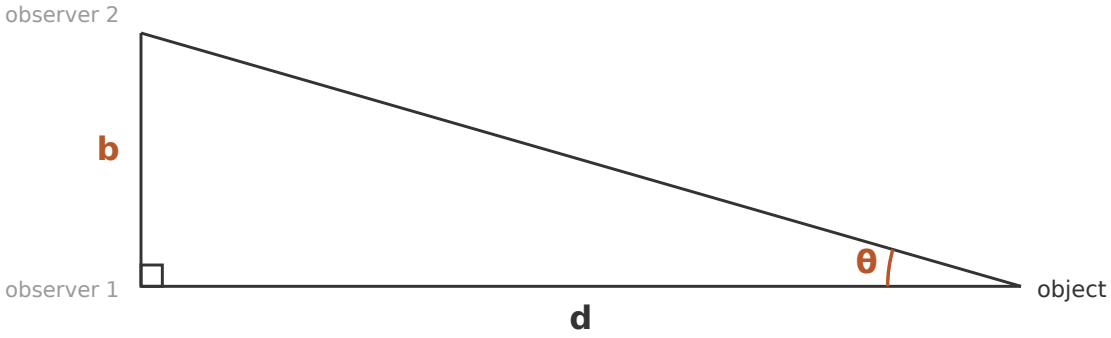
Panel 1 — the geometry (schematic side view; not to scale; angles hugely exaggerated)



How to read it: both observers look at the same Venus. Because they stand in different places, their sight lines cross at Venus and land on different spots on the Sun. The Sun is also at a finite distance, so it shifts a little too; what observers actually measure is Venus’s shift RELATIVE to the Sun (Venus’s parallax minus the Sun’s own parallax). In 1761, that shift showed up as differences in contact timings. Schematic note: London and Cape Town are NOT opposite ends of a diameter and are not placed from a real map here; the useful baseline is only the part of their separation perpendicular to the line of sight, and it changes hour by hour as Earth rotates.

Panel 2 — turning an angle into a distance

Basic trigonometry; round, historical-style numbers for illustration (not the 1761 London–Cape Town measurement)



Right triangle, b perpendicular to d, θ at the far object:

d = b / tan θ (exact for this one-sided triangle)
Symmetric case (b split evenly about the sight line):
d = b / [2 tan(θ/2)] ≈ b / θ (θ in radians)

Solar inset — the classic textbook triangle

Solar horizontal parallax p☉ = the angle one Earth RADIUS (≈ 3,959 mi, ~4,000 rounded) spans as seen from the Sun.

Using p☉ = 8.5″ = 8.5 / 206,265 rad ≈ 0.0000412 rad:
d☉ = R / tan p☉ ≈ 3,959 / 0.0000412 ≈ 96 million mi
(≈ 97 million mi with R rounded to 4,000)

Historical illustration only. 8.5″ was an assumed working value (e.g., James Short, 1761) — NOT Halley's 1716 figure (he assumed 12.5″). Modern value ≈ 8.794″ → 1 AU ≈ 93.0 million mi.

From the Sun to Venus — and everything else (Kepler's third law)

$P^2 \propto a^3$. From the known orbital periods (Venus 224.7 days, Earth 365.25 days): a_Venus / a_Earth = (224.7/365.25)^(2/3) ≈ 0.723.
At inferior conjunction, with circular orbits as an approximation: Earth–Venus ≈ (1 − 0.723) × Earth–Sun ≈ 0.277 × 96 M mi ≈ 27 million mi.
At that distance one Earth radius spans ≈ 30.7″ from Venus; Venus's shift relative to the Sun ≈ 30.7″ − 8.5″ ≈ 22.2″ (double for a full diameter).
Kepler gives ratios only; one measured distance sets the scale for all planets. Real 1761 analyses used full orbital geometry, not circles.

Historical footnote

In 1761 Cape Town and Greenwich could compare only the egress (end), not a whole transit. Short found 435 s observed vs 439 s predicted for 8.5″, implying ≈ 8.42″. Illustrative numbers above are not those observations.

Sources

Halley (1716), English tr.: eclipse.gsfc.nasa.gov/transit/HalleyParallax.html · Short (1761), Phil. Trans.: doi.org/10.1098/rstl.1761.0102
NASA Venus fact sheet (orbit, period): nssdc.gsfc.nasa.gov/planetary/factsheet/venusfact.html